

K Madhumitha

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ABOUT ME

Computer Science student with a strong interest in full-stack development, artificial intelligence, and computer vision. Experienced in building AI-powered applications and implementing machine learning models using Python, React, SQL, and modern web technologies. Seeking opportunities to contribute to innovative and impactful projects.

PROFESSIONAL EXPERIENCE

Research Intern, Indian Space Research Organization (ISRO) 04/2026 – 05/2026
Bangalore, India

- Researched vision-based satellite pose estimation techniques and analyzed multiple deep learning research papers focused on computer vision and object detection.
- Implemented, trained, validated, and compared YOLOv8 and RF-DETR models on satellite datasets, achieving comparable accuracy through performance metrics evaluation.

Artificial Intelligence Intern, Pinnacle Labs 06/2025 – 07/2025
Bangalore, India

- Developed two full-stack AI applications: a grammar correction tool (Python, Flask, Hugging Face) and a voice-enabled translator (Google Gemini API).
- Designed responsive UIs for both projects with Tailwind CSS and collaborated in a remote, agile environment using Git for version control.

SKILLS

Languages

JavaScript (ES6+), SQL, HTML5, CSS3, Java, Python

Frameworks / Libraries

React.js, Node.js, Express.js, YOLOv8, RF-DETR, OpenCV

Databases

Supabase, MySQL, MongoDB

Developer Tools

Git, GitHub, Netlify, Canva, Jupyter Notebook

PROJECTS

NovelNest 📁, (Personal project) 12/2025 – 01/2026

- Built a full-stack multi-user content platform with secure authentication, role-based access control, approval workflows, and real-time cloud data management using Supabase.
- Developed an interactive reading experience featuring PDF viewing, bookmarking, progress tracking, filtering, and a moderated commenting system.
- Tech Stack: React.js, Tailwind CSS, Supabase, Row Level Security (RLS).

Expense Tracker 📁, (Personal Project) 09/2025 – 11/2025

- Developed a secure full-stack expense management application with multi-account transaction tracking, recurring payment automation, budgeting alerts, and interactive financial analytics.
- Implemented advanced finance management features including searchable transaction history, pagination, CSV import/export, and dynamic Chart.js visualizations.
- Tech Stack: MERN Stack, JWT Authentication, Chart.js, Tailwind CSS.

CERTIFICATIONS

Data Science using Python (Infosys Springboard)

Frontend Development (Infosys Springboard)

Programming in Java 📁 ((Elite) - NPTEL)

HACKATHONS

Suprathon 📁, SuPrazo Technologies & CodeElevate Academy 30/07/2025

- Built a Resume Builder web app using React and Tailwind CSS, deployed on Netlify, enabling users to quickly create, edit, and customize professional resumes with a responsive and scalable design.